Chapter 4

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

4.1 Introduction and Scope

?? established, on a single fully characterised *Super Yellow*, *Hybrane*, and LiOTf device, that a three-component polymer-electrolyte composite supports the full set of memristive functions in one two-terminal platform. The natural next question — and the subject of this chapter — is how the *dynamics* of these devices respond to deliberate changes in the composite, so that the volatile, fading-memory behaviour can be treated as something to be engineered rather than merely observed.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next, and are introduced here as a named set. They are (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before read-out, which reports how a written state fades. These three measurements are the shared, abundant basis of the comparative work. The richer single-device synaptic protocols of **??** — excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity — are *not* repeated across the families studied here; where they are invoked, they are used only as prior evidence from the Chapter **??** device and are labelled as such.

Claim discipline. A central feature of this chapter is that its results fall into two evidential tiers. The **composition** axis — variation of the ion-transport polymer and salt mass fractions in the *Super Yellow*/PEO/LiOTf system — is supported by a replicated design (typically two to four nominally identical devices per composition cell) and is presented as a *quantitative* result. The **chemistry** axis — variation of the ion-transport host, the salt anion, and the alkali cation — is supported, after data-quality screening, by at most one or two devices per matched cell; it is therefore presented as an *illustrative* landscape, with sample sizes stated explicitly and no claim of statistical power. The **electrode** (active silver versus inert gold) is a further illustrative axis, available only at the lead composition and reported on that footing (section 4.8). Finally, a **methodological** result establishes that the *drive protocol* itself sets the apparent fading-memory timescale, which both explains why the chemistry comparisons must be read cautiously and provides a design rule for any future comparative study.

4.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of **??**: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction area of $8.25 \times 10^{-2}\,\mathrm{cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

Two design axes are varied. The *composition* axis varies the mass fraction of the ion-transport polymer poly(ethylene oxide) (PEO) and of the lithium triflate salt (LiOTf) relative to the semiconductor. The *chemistry* axis substitutes the ion-transport host, the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+, Na^+, K^+, as the triflate or TFSI salt). The comparative work replaces the hyperbranched polyester-amide *Hybrane* host of the proof-of-concept device (**??**) with two hosts used here at the same level: linear PEO and the hyperbranched polyether trimethylolpropane ethoxylate (TMPE), so that the host axis contrasts a linear against a branched polyether at matched chemistry. Every device reported *quantitatively* uses the silver electrode. A separate corpus of gold-electrode devices also exists, entirely at the lead composition; rather than discard it, the electrode is treated as a third, illustrative axis in its own right (section 4.8), where it both contrasts an active against an inert electrode and provides an independent-electrode check on the chemistry comparisons.

Film thickness is a controlled covariate, not a confound. Because adding more ion-transport polymer thickens the solution-processed film, the spin-coating speed was, in most batches, raised for the higher-PEO cells specifically to partially equalise thickness across the grid (spin speeds span 1500 rpm–3500 rpm); this compensation was deliberate but not applied uniformly to every comparison batch. The natural question is therefore whether the composition trends below are in fact thickness effects. They are not, and the point is settled directly against the profilometry record (the silver composition spine, $n = 30$; fig. 4.1). The compensation is real but incomplete, so a residual covariation survives — median film thickness rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2 (Pearson $r \approx 0.68$ between PEO fraction and thickness). Four lines of evidence show that thickness is nonetheless not the operative variable: the reported metrics are dimensionless ratios and timescales in which a geometric thickness effect cancels; the activation voltage is flat (2.2 V–2.4 V, $r \approx -0.06$) across the full 2.6× thickness range; a 38% within-cell thickness spread leaves the fading-memory time essentially unchanged; and, decisively, the partial correlation of thickness with $\log t_{1/2}$ controlling for PEO vanishes ($\approx +0.05$) while the composition effect survives controlling for thickness (≈ -0.42). Thickness covaries with PEO as a downstream consequence of polymer loading, but the dynamics track composition. The four arguments with their full statistics, and a column-by-column audit confirming that every other fabrication variable is held constant or crossed with composition rather than confounded with it, are given in Appendix C.

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly — and, importantly, they are *not* identical across the three measurements (table 4.1). Hysteresis uses a triangular voltage sweep. The potentiation (N-pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V) — a point

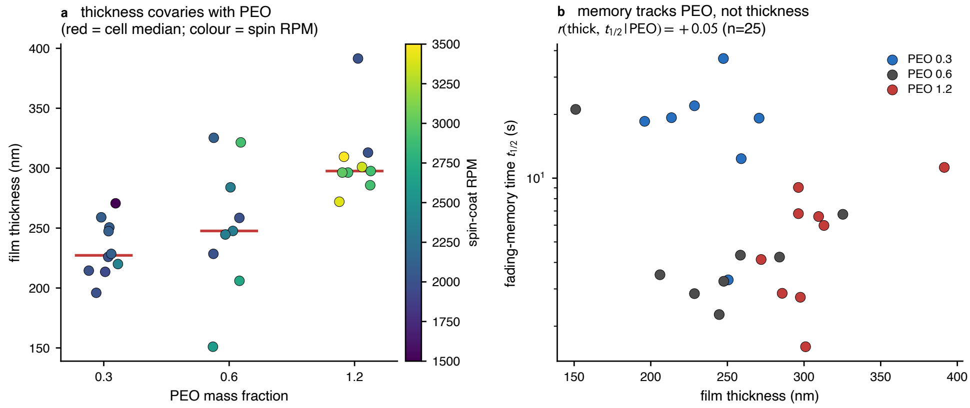

Figure 4.1: Film thickness covaries with composition but does not drive the dynamics (silver composition spine, $n = 30$). (a) Film thickness rises with PEO fraction (red bars are cell medians); the spin-coating speed (colour) was raised for higher-PEO cells to partially equalise thickness, but the compensation is incomplete. (b) The fading-memory time $t_{1/2}$ is set by composition (PEO colour), not by thickness (horizontal axis): within each PEO band the timescale is flat against thickness, and the partial correlation of thickness with $t_{1/2}$ controlling for PEO is ≈ 0. Data from `DEVICES_PROFILOMETRY_STATS.csv`; see `handouts/14_thickness_rpm_confound_audit.md`.

that matters when the two are later composed (section 4.10, ??). The read voltage is above the ionic activation threshold established in ??, a point returned to in section 4.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 4.6).

4.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth of the conductance ratio with pulse number, by the peak ratio, and by

Table 4.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Crucially, potentiation and depotentiation differ in write amplitude, so the chapter never pools them and **??** composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$, 0.104 s
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the comparative archive is refitted with the more general Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\text{wait}}) = \Delta G_0 \, \exp\left[-\left(\frac{t_{\text{wait}}}{\tau}\right)^{\beta}\right] + \Delta G_{\infty}, \tag{4.1}$$

with characteristic relaxation time τ, stretch exponent $0 < \beta \leq 1$, initial change ΔG_0, and residual ΔG_{∞} [1]. Two practical points follow from the data. First, the eight-to-thirteen-point decay curves often under-constrain β and τ jointly; the fitted τ is therefore reported only where the fit is well identified, and a robust, model-free half-enhancement time $t_{1/2}$ (the delay at which the conductance enhancement has fallen to half its initial value) is used as the primary timescale. Where both are available they agree. Both are *apparent* timescales: they depend not only on the material but on the amplitude used to write and read the state, a methodological point established in section 4.6 that disciplines every cross-protocol timescale comparison below — and the reason the composition spine and the chemistry landscape are each measured under a single fixed protocol. Second, fitted β values lie predominantly in the range 0.6–0.9, i.e. genuinely stretched relaxation, consistent with the heterogeneous distribution of activation barriers invoked in **??**.

Data-quality control and aggregation. Two screening layers are applied before any number is reported: erratic or non-physical measurements are flagged and excluded, and a per-device visual review stores per-curve verdicts (used as measured, cleaned, or discarded) in a curation registry applied automatically by the analysis scripts. All timescales are recomputed after this screening, because the instrument-side fit is not robust to anomalous points. Summaries then follow a deliberately conservative curve→pixel→device→cell ladder of medians, and the representative curves of fig. 4.2 are drawn by script from devices near their cell medians rather than hand-picked. The full raw-to-summary provenance trail, the aggregation ladder, and the stretched-exponential identification thresholds are documented in Appendix C.

4.4 Composition Control of the Dynamics

The quantitative core of this chapter is the *Super Yellow*/PEO/LiOTf composition grid, in which the PEO mass fraction is varied over 0.3, 0.6 and 1.2 and the salt mass fraction over 0.045, 0.09 and 0.18, with two to four nominally identical silver-electrode devices per cell. Each measurement was acquired under its own fixed protocol (table 4.1); the composition contrasts below are therefore always drawn *within* a single measurement, never by pooling the 3 V potentiation with the 4 V depotentiation. Before the grid is summarised, fig. 4.2 shows what the three underlying measurements actually look like on representative devices, so that the on–off ratio, the growth exponent, the turnover, and the fading-memory time can be read off real curves before they are reduced

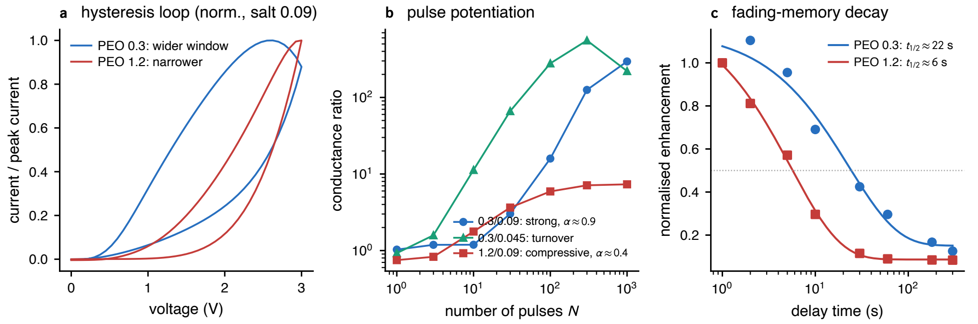

Figure 4.2: Representative raw curves of the three common measurements, on devices chosen near their cell medians (*Super Yellow*/PEO/LiOTf/Ag). (a) Hysteresis loops, each normalised to its own peak current so the comparison is the loop *openness* (the switching window) rather than absolute conductance: the low-PEO device (PEO 0.3, blue) opens a wider window than the high-PEO one (PEO 1.2, red), which carries more current but switches over a narrower range. (b) Pulse potentiation (conductance ratio vs pulse number, log–log): a strong near-linear accumulation ($\alpha \approx 0.9$), a low-salt cell that *turns over* and then declines, and a weak, compressive high-PEO response ($\alpha \approx 0.4$). (c) Fading-memory decay (enhancement vs delay, normalised to the first read) with stretched-exponential fits and the half-enhancement level dotted: the low-PEO device crosses $t_{1/2}$ far later ($\approx 22\,\text{s}$) than the high-PEO one ($\approx 6\,\text{s}$). Curves drawn directly from the DATABASE by `scripts/ch3_figures.py`.

to per-cell numbers. Three trends, mutually consistent and each visible across the grid, define the result.

First, the steady-state switching window contracts as the PEO fraction rises. At fixed salt mass fraction 0.09, the median on–off ratio falls from ≈ 3.3 at PEO 0.3 to ≈ 2.4 at PEO 1.2, and the normalised loop area falls correspondingly from ≈ 0.33 to ≈ 0.18; the activation voltage is, by contrast, essentially flat across the whole grid (2.2 V–2.4 V), indicating that composition tunes how *much* the conductance moves rather than the voltage at which it begins to move. The widest window in the grid sits in the low-PEO row, with the largest loop area at the low-salt corner (PEO 0.3, salt 0.045: area ≈ 0.44); the per-sweep on–off ratio is strongly

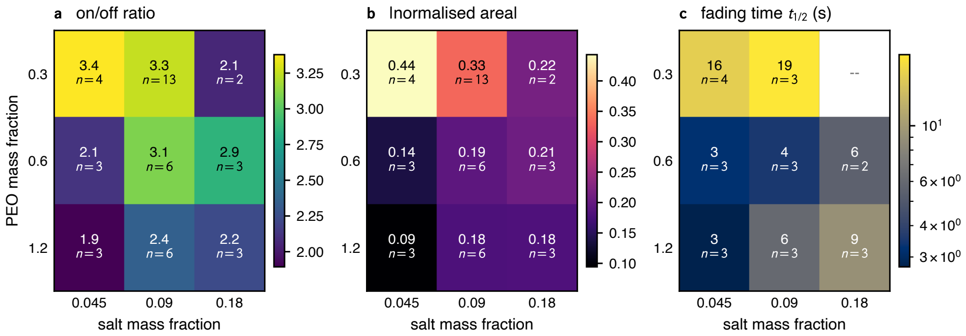

Figure 4.3: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. The switching window (a, b) and the fading-memory time (c) both decrease as the PEO mass fraction increases at fixed salt content.

right-skewed there (cell median ≈ 3.4 but individual sweeps reaching ~ 40), so robust cell medians rather than means are reported throughout to avoid letting a few outlying sweeps define a cell.

Second, potentiation weakens, and changes shape, as the composition is varied (fig. 4.5). Two axes act distinctly. The PEO fraction controls the *strength* of pulse integration: the per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content — most cleanly at the lowest salt, where it drops from $\alpha \approx 1.1$ (near-linear accumulation) at PEO 0.3 to $\alpha \approx 0.3$ (strongly compressive) at PEO 1.2 — and the peak conductance ratio, i.e. the usable dynamic range, is largest in the low-PEO, low-salt corner ($\approx 480\times$ at PEO 0.3/salt 0.045) and falls to a few-fold at high PEO. The salt fraction, by contrast, controls *how far* the response can be driven before it saturates: at the lowest salt (0.045) every cell *turns over* — the conductance ratio peaks and then declines — by $N \sim 100$–300 pulses, whereas at the highest salt (0.18) the response grows monotonically through the largest counts tested ($N \approx 1000$), with salt 0.09 intermediate. Two caveats are kept in view: the α trend is noisier at the higher salts, where the sparsely sampled PEO 0.3/salt 0.18

cell ($n = 2$, curation-salvaged) departs from it; and the turnover sets a practical ceiling on the usable pulse count. Read constructively, these responses are a composition-tunable, predominantly compressive and (in the low-salt cells) non-monotone integration of the input pulse train — the nonlinear input encoding that **??** builds on.

Third, and most important for the rest of the thesis, the fading-memory timescale is composition-tunable over roughly 2 s–20 s. The longest retention occurs in the low-PEO row (at PEO 0.3, $t_{1/2} \approx 19\,\mathrm{s}$ at salt 0.09, with an identified $\tau \approx 25\,\mathrm{s}$); increasing the PEO fraction to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-salvaged device at the still-lower PEO 0.15 level ($t_{1/2} \approx 22\,\mathrm{s}$) is consistent with this trend, though it lies outside the replicated grid and is not counted toward it. The same ordering is reproduced by the instrument-side single-exponential time constant computed independently, which provides an internal cross-check on the trend. The device-to-device scatter within each cell is substantial and is not treated here as mere noise (fig. 4.4): the spread of timescales *across* the grid, together with the within-cell variability, constitutes the heterogeneous bank of fading-memory elements that **??** exploits. The same composition axis is visible in the potentiation strength: the growth exponent α falls monotonically with PEO fraction (cell medians $\approx 0.9 \to 0.7 \to 0.35$), again with real within-cell spread rather than a single value per composition.

Taken together, these observations support a single physical reading: increasing the proportion of the ion-transport phase produces a weaker, faster-relaxing memristive response, while the salt content governs how far that response can be pushed before it saturates. This is the quantitative spine of the chapter, and the one result that rests on replicated devices; the per-cell decay and potentiation values behind the heatmaps are tabulated in Appendix C. The mechanism it implies — that a more ion-rich matrix relaxes a displaced ionic profile more readily — makes an independent,

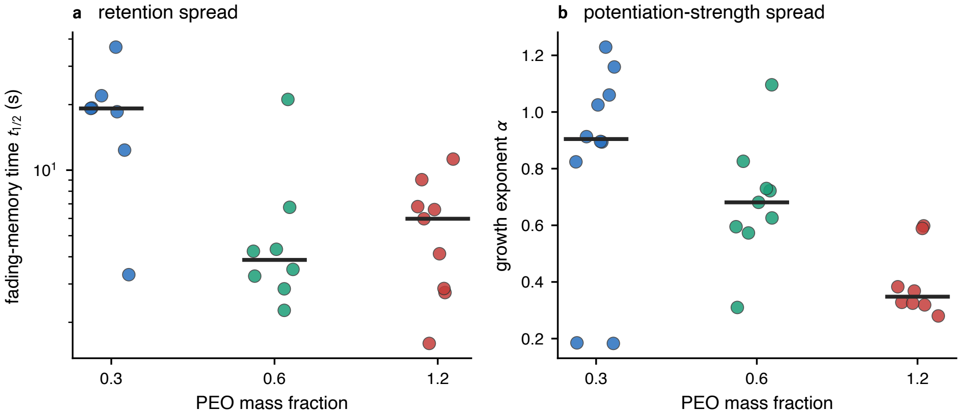

Figure 4.4: The cell medians of fig. 4.3 are summaries of genuinely spread distributions, not tight clusters (one point per device, *Super Yellow*/PEO/LiOTf/Ag; black bar is the cell median). (a) Fading-memory time $t_{1/2}$ and (b) potentiation growth exponent α, each against PEO fraction. The PEO fraction sets the broad level, but within every composition cell the device-to-device variation is comparable to the steps between cells — the heterogeneity that **??** treats as a resource rather than averaging away. Per-device values from `handouts/ch4_decay_fits.csv` and `ch4_pulse_descriptors.csv`.

equilibrium prediction about the ionic resistance of these films; that prediction is tested directly by impedance spectroscopy in section 4.7.

4.5 Chemical-Tuning Landscape

Beyond composition, the identity of the electrolyte components shifts the dynamics further. These comparisons are reported as an illustrative landscape: each matched cell (fixed composition, electrode, and protocol) contains at most one or two screened devices, so the magnitudes below are indicative and the sample size is given in every case. They are complemented here by structural and optical probes on *separate* sample sets — blend films and scratched powders at the lead composition (0.3/0.09)

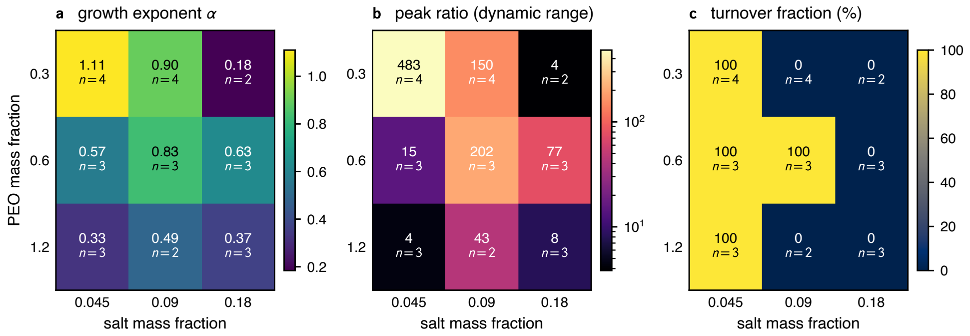

Figure 4.5: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α, (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of devices whose response turns over within the measured range ($N \leq 1000$). Increasing the PEO fraction lowers α and the dynamic range; increasing the salt fraction suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the device count n annotated in each cell.

— examined by ATR-FTIR, X-ray diffraction, and UV-Vis absorption, which probe ion association, host backbone order, crystallinity, and the semiconductor's electronic gap directly, independently of how a device is driven (figs. 4.6 and 4.8). These too are single specimens per chemistry and are read illustratively.

Host. Substituting the linear polyether PEO by the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf: $t_{1/2} \approx 20\,\mathrm{s}$–$25\,\mathrm{s}$, three screened silver devices; TMPE/LiOTf: $\approx 4\,\mathrm{s}$, one screened silver device with a clear decay). This is the best-replicated of the chemistry comparisons and the clearest single chemistry effect, although the TMPE side rests on a single clean device. A caveat of provenance applies: the cleanest test would have been the lithium pair from a single designed host$\times$cation batch, but the PEO/LiOTf corner of that batch was a broken device and had to be discarded, so the comparison instead pairs

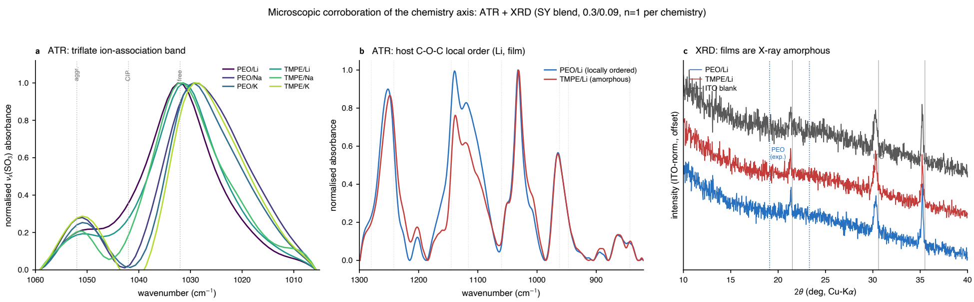

Figure 4.6: Microscopic corroboration of the chemistry axis (electrode-free *Super Yellow*-blend films/powders, 0.3/0.09, one specimen per chemistry). (a) ATR-FTIR triflate ion-association band $\nu_s(SO_3)$, whose position reports the ion-association state (spectroscopically free near 1032 cm^{-1}, contact ion pair near 1042 cm^{-1}, aggregate above; reference positions dotted [2]): it sits at 1029 cm^{-1}–1032 cm^{-1} in *every* host×cation sample with no cation-resolved shift — microscopically consistent with the cation not being a clean retention lever. (b) ATR-FTIR C–O–C envelope: the linear PEO host (blue) shows a resolved multiplet, i.e. retained *local* (conformational) ether-backbone order, whereas the hyperbranched TMPE host (red) collapses it into a single broad, amorphous band [2]. (c) X-ray diffraction of PEO/Li and TMPE/Li with a bare-ITO blank: all patterns show only the crystalline ITO substrate reflections (grey lines; 21.5°, 30.6° and 35.5°), with no sharp PEO crystalline doublet (blue dotted; 19.1° and 23.3°) and no crystalline salt phase — the composite films are X-ray amorphous, so PEO's order in (b) is short-range, not bulk crystallinity [2, 3]. ATR spectra normalised after a local/rubber-band baseline and XRD normalised to the ITO peak; intensities are not comparable across specimens (mixed film/powder sampling and thin films), so only band *positions* and *shapes* are read. Analysis in `scripts/ch3_iratr.py` and `scripts/ch3_xrd.py`; assessments in `handouts/16_iratr_assessment.md` and `17_xrd_assessment.md`.

the TMPE device against the nearest-matched silver PEO/LiOTf devices (same operator, fresh semiconductor, glovebox), which differ in fabrication date. An independent microscopic correlate of the host substitution is visible in the polyether backbone itself, and at two length scales. In ATR-FTIR the linear PEO film shows a resolved C–O–C stretching multiplet, with the weak order-sensitive marker bands near 843 cm^{-1}, 962 cm^{-1} and 1242 cm^{-1}, indicating retained *local* (conformational) order of the ether

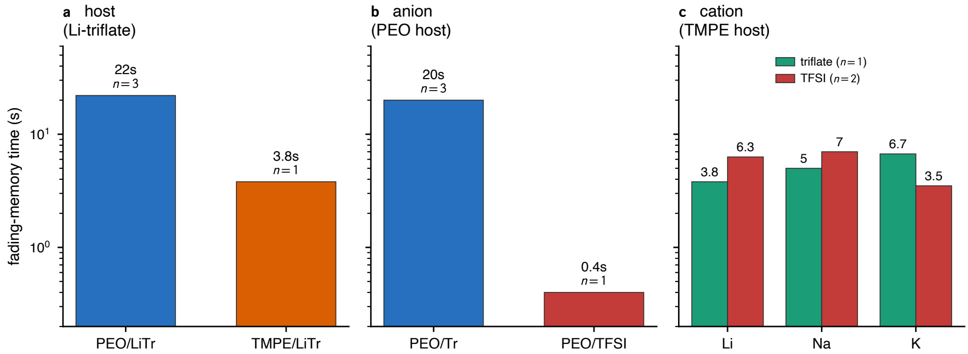

Figure 4.7: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: the order *inverts* with the anion (potassium longest in triflate, shortest in TFSI), so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

backbone, whereas the hyperbranched TMPE film collapses these into the single broad band of an amorphous polyether (fig. 4.6b) [2]. At the longer length scale of crystallites, however, X-ray diffraction on the same sample set finds *neither* host crystalline in the composite: every pattern is dominated by the ITO substrate (confirmed against a bare-substrate blank), with no sharp PEO crystalline reflections and no segregated crystalline salt phase (fig. 4.6c), so the films are X-ray amorphous and PEO retains short-range order without forming long-range crystallites [2, 3]. A host that is locally more ordered, even while globally amorphous, set against a fully amorphous TMPE, is consistent with the lower segmental mobility and slower ionic relaxation that the *longer* PEO fading-memory time would require: in polyether electrolytes ion transport is carried by segmental motion of the host above its glass transition, so the relaxation of a displaced ionic profile is rate-limited by that segmental mobility [4–6]. With one specimen per host this is offered as a structural correlate rather than a demonstrated cause. The host ordering is moreover not specific to the silver electrode: on the independent gold-electrode corpus at the same composition, PEO again retains far longer than TMPE and supports a markedly wider switching window and stronger potentiation (section 4.8), so the host effect survives a change of electrode. That the salt forms no crystalline phase in any chemistry additionally supports reading these blends as homogeneous, salt-dissolved ion-conducting media.

Anion. Replacing triflate by the more charge-delocalised TFSI anion, in the PEO host, shortens the fading-memory time dramatically — from $\approx$ 20 s (triflate) to of order 1 s (TFSI). The direction is the one the electrolyte literature would anticipate: the delocalised TFSI charge weakens ion pairing and lowers the salt lattice energy, so that TFSI salts dissolve and dissociate more freely and plasticise the polyether host more strongly than triflate, raising ionic mobility [7, 8] — all of which should speed the relaxation of a driven ionic profile. The TFSI decays additionally tend to a sharper, more abrupt (compressed) shape rather than the stretched form of the triflate devices, suggesting a qualitatively different relaxation-time distribution. This is the most weakly controlled of the chemistry comparisons: the single PEO/TFSI device belongs to a later fabrication generation than the triflate devices and differs from them not only in the anion but also in spin speed and in whether the salt was filtered before use. The effect is large enough, and the read-out metrics dimensionless enough (section 4.2), that these accompanying differences are unlikely to account for a twenty-fold change, but the comparison is reported as illustrative for this reason as much as for its sample size.

Cation. The original motivating hypothesis — that the alkali cation would order the fading-memory timescale as $Li^+ > Na^+ > K^+$, by analogy with the hard–soft acid–base argument of **??** [9] — is *not* supported by the data. The decisive evidence is a pair of cation series measured in the *same* host (TMPE), at the same composition, electrode, and protocol, differing only in the salt anion (the cation panel of fig. 4.7). With triflate, the timescale *increases* along the series — $Li^+ \approx 3.8$ s, $Na^+ \approx 5.0$ s, $K^+ \approx 6.7$ s (one screened device per cation, all $R^2 \geq 0.93$) — so potassium retains *longest*, the reverse of the hypothesised order. With TFSI, the same three cations invert: potassium is now *shortest* (≈ 3.5 s) and lithium and sodium are together and longer (≈ 6 s–7 s, two devices per cation). A single change of anion thus flips which cation persists longest, while in each series the within-cation device-to-device variation is comparable to the between-cation differences. The conclusion is that, in this archive, cation identity does not produce a robust, host- and anion-independent retention ordering: any apparent ordering (such as potassium being shortest in the TFSI family, which is consistent with the weakest cation–oxygen coordination) is contingent on the rest of the chemistry. A properly powered test of the cation hypothesis would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell, and is identified as future work in section 4.10.

This cation null is not an artefact of the driven measurement: three independent probes reach it from three directions. The first is ion association. The triflate symmetric SO_3 stretch $\nu_s(SO_3)$ sits at 1029 cm^{-1}–1032 cm^{-1} in *all six* host×cation films and powders, with a spread at or below the instrumental resolution and no ordering by cation (fig. 4.6a); the band lies at the spectroscopically-free / contact-ion-pair boundary in every case. At the level of ion pairing, that is, the cation leaves no host-independent fingerprint — the same conclusion the driven retention reaches, by an independent probe. The second is a change of electrode: on the gold-electrode TMPE/triflate cation series at the same composition, the steady-state switching window is flat across Li^+, Na^+, and K^+ (on–off ratio ≈ 1.3–1.5 with no monotone ordering; section 4.8), so

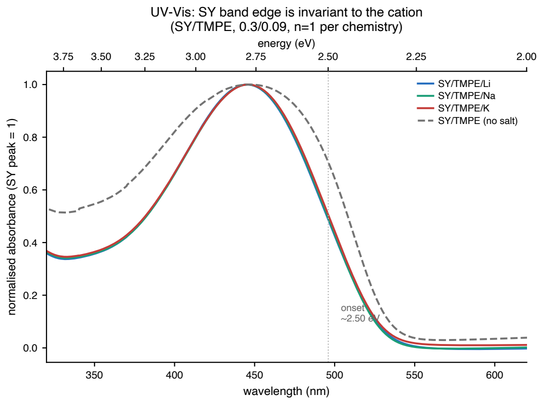

Figure 4.8: UV-Vis absorption of *Super Yellow*/TMPE blend films (lead composition 0.3/0.09; Au-electrode 2025 generation; one specimen per chemistry), normalised to the *Super Yellow* π–π^* peak. The peak (≈ 446 nm) and the absorption onset (≈ 2.50 eV) are superimposed for Li^+, Na^+, and K^+; the salt-free control (dashed) coincides at the band edge (its differing blue side and lower absolute absorbance are thickness/scattering, not electronic). The cation does not shift the semiconductor's optical gap. Only band *positions* are read (absorbance magnitudes are thickness-confounded); analysis in `scripts/ch3_uvvis.py`, assessment in `handouts/18_uvvis_assessment.md`.

a second electrode and a second observable return the same absence of a clean cation ordering. The third probe is the electronic structure: UV-Vis absorption of *Super Yellow*/TMPE blend films across the cation series (a separate, later Au-electrode generation) places the *Super Yellow* π–π^* absorption peak of this poly(phenylene-vinylene) derivative [10] at 446 nm with an onset of ≈ 2.50 eV *identically* for Li^+, Na^+, and K^+ (a salt-free control matches at the band edge), with no new sub-gap feature (fig. 4.8). The cation thus perturbs neither the ion-association state nor the semiconductor's optical gap; consistent with the design premise of the chapter, the electrolyte chemistry tunes the ionic dynamics while leaving the *Super Yellow* electronic backbone fixed.

The hierarchy that emerges from this landscape is therefore that the host and the anion shift the fading-memory time more strongly, and more

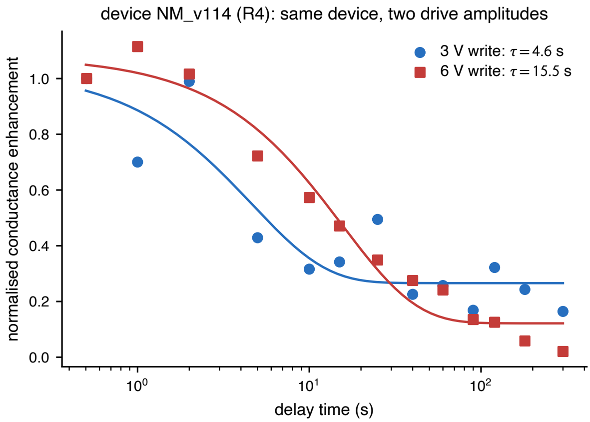

Figure 4.9: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time.

legibly, than the cation does.

4.6 The Drive Protocol Sets the Timescale

A methodological result underlies, and disciplines, every comparison above. Because the read voltage used to interrogate the state is above the ionic activation threshold, and because the write amplitude controls how far the ionic profile is driven, the *drive protocol* itself influences the apparent fading-memory time. The point is made cleanly by a single device measured under both protocols: its fitted relaxation time increases from $\approx 4.6\,\mathrm{s}$ under the 3 V-write/1.5 V-read protocol to $\approx 15.5\,\mathrm{s}$ under the 6 V-write/3 V-read protocol — a roughly three-fold change in the measured timescale with no change in material, only in drive (fig. 4.9).

The mechanistic reading is that a larger write amplitude displaces more of the ionic charge (and, at the largest amplitudes, may drive electrode-level electrochemical processes), producing a deeper, longer-lived written state, while the supra-threshold read perturbs the state during interrogation.

Two consequences follow. First, the composition result of section 4.4 is unaffected, because it was obtained entirely within a single, fixed protocol; only the chemistry comparisons, which unavoidably span device generations, are exposed to this confound, and they are reported accordingly. Second, the result is a positive design rule for the field: the volatile timescale of these devices is not a fixed material constant but is co-determined by how the device is driven, and any meaningful cross-device or cross-chemistry comparison of retention must hold the write/read protocol, the electrode, and the composition fixed.

4.7 Microscopic Corroboration: Impedance Spectroscopy

Every result so far rests on the *driven* response — pulses and decays. Small-signal impedance spectroscopy offers an independent test of the mechanism proposed for the composition trend: if increasing the PEO fraction shortens the fading-memory time because a more ion-rich matrix lets the displaced ionic profile relax more readily, then the *equilibrium* ionic resistance should fall along the same composition axis. The archive contains such a measurement, acquired on the same devices: 0 V-DC spectra (40 mV AC, 1 Hz–1×10^6 Hz) on 26 silver-electrode *Super Yellow*/PEO/LiOTf devices spanning the composition grid.

Each spectrum is summarised in two deliberately separated ways. The first is *model-free*: the real part of the impedance at the apex of the main Nyquist semicircle, $Z_{\mathrm{real}}^{\mathrm{apex}}$, which fixes the resistance scale of the spectrum without committing to any circuit. The second is *model-based*: the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [11], reported only as corroboration of the model-free quantity because the per-spectrum

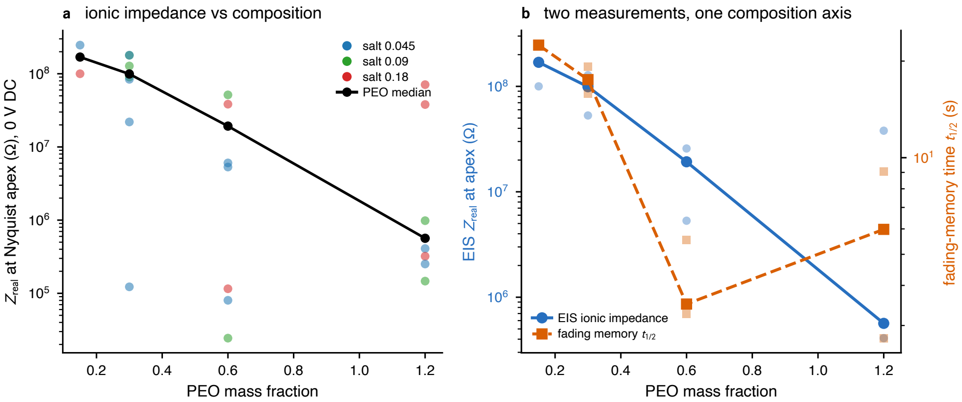

Figure 4.10: Equilibrium (0 V-DC) impedance spectroscopy corroborates the composition control of the dynamics (silver *Super Yellow*/PEO/LiOTf grid). (a) The impedance resistance scale — here the model-free real impedance at the Nyquist apex — falls by more than three orders of magnitude as the PEO fraction rises (points are devices, coloured by salt; the black line is the PEO median). (b) The equilibrium impedance scale (left axis) and the *independently* measured fading-memory time $t_{1/2}$ (right axis) decline along the *same* PEO composition axis (cell-level points with per-PEO median trends): the two are co-driven by composition, not independent knobs. Analysis in `scripts/ch3_eis.py`; per-cell values in `handouts/ch3_eis_by_cell.csv`.

circuit fits are genuinely ill-conditioned. The circuit topology, its two automated fits, and the per-cell values of both observables are detailed in Appendix C.

Both observables fall steeply and monotonically with PEO fraction (fig. 4.10a). The model-free $Z_{\mathrm{real}}^{\mathrm{apex}}$ drops by more than three orders of magnitude across the grid — cell medians from $\sim 10^8\,\Omega$ at PEO 0.15–0.3 to $\sim 10^5\,\Omega$ at PEO 1.2 — with a rank correlation against PEO of -0.69 per device ($n = 26$) and -0.83 on cell medians; the fitted R_{ion} gives the same picture (-0.84 on cell medians). This is exactly the direction the mechanism requires: more ion-transport polymer, lower ionic resistance.

Crucially, the EIS resistance scale and the *independently* measured fading-memory time fall along the *same* composition axis (fig. 4.10b): both decline together as the PEO fraction rises, and across cells $\log Z_{\mathrm{real}}^{\mathrm{apex}}$ covaries with $\log t_{1/2}$ at $r \approx +0.74$ (+0.61 per device, $n = 23$). Three qualifications bound this reading:

- The PEO fraction is the *common cause* of both quantities, so this correlation confirms that the same composition axis moves the equilibrium ionic resistance and the driven retention together — it is not evidence of a second, independent factor.
- It is the resistance *scale* that carries the composition signal, not the EIS relaxation *frequency*: the apex frequency corresponds to a sub-second RC time and does *not* track the seconds-long fading-memory time, consistent with impedance spectroscopy reporting the small-signal ionic-transport bottleneck rather than the slow relaxation that governs forgetting.
- The effect is not a film-thickness artefact — $Z_{\mathrm{real}}^{\mathrm{apex}}$ spans $\sim 10^4\times$ while the film thickness spans only $2.6\times$, and the PEO dependence survives controlling for thickness (partial correlation -0.60; section 4.2).

Impedance spectroscopy thus supplies, for the ionic-resistance half of the mechanistic picture below, the independent microscopic support that the driven measurements alone cannot.

4.8 Electrode Dependence: Active Silver versus Inert Gold

Every result so far uses the silver top electrode. The archive also holds a gold-electrode corpus (50 devices), and rather than set it aside it is read here as a third, illustrative axis. One structural fact governs how it can be read: *every* gold device sits at the single lead composition (PEO or TMPE host, 0.3/0.09), so the gold corpus is a chemistry-axis set at fixed composition and carries *no* information about the composition spine of section 4.4, which therefore remains a silver-only result. What the gold corpus can do is twofold: contrast an electrochemically active electrode (silver) against a noble, inert one (gold) at matched chemistry, and provide an independent-electrode check on the chemistry comparisons of section 4.5.

At matched chemistry and composition the electrode shifts all three behavioural families, and does so coherently (fig. 4.11). In the lead PEO/LiOTf cell, replacing silver by gold narrows the steady-state switching window (median on–off ratio 3.3 $\rightarrow$ 2.0, normalised loop area 0.33 $\rightarrow$ 0.12), weakens and flattens potentiation (peak conductance ratio $\approx$ 124$\times$ $\rightarrow$ 29$\times$, log–log slope $\alpha \approx 0.9 \rightarrow 0.45$), and raises the activation voltage ($\approx$ 2.4 V–2.7 V) — yet *lengthens* the fading-memory time, from $t_{1/2} \approx 19$ s on silver (three devices) to ≈ 87 s on the two clean gold devices ($R^2 \approx 0.97$–0.99, $\beta \approx 0.6$). The same window-and-potentiation contraction holds in the TMPE/LiOTf cell (on–off 3.8 $\rightarrow$ 1.3; peak 54$\times$ $\rightarrow$ 1.9$\times$). The per-cell metrics for every chemistry on both electrodes, including the host-free and salt-free negative controls, are tabulated in Appendix C.

The natural reading is the textbook contrast between an active and an inert electrode. Silver is electrochemically active — mobile Ag^+ can be injected and electrodeposited, the basis of electrochemical-metallisation switching — and so contributes a large but *volatile* component to the response: a wider window, a stronger and steeper potentiation, a lower threshold, and a fast-relaxing written state [12, 13]. Gold is noble and inert, so the switching is carried by the polymer-electrolyte ionic mechanism alone: smaller in amplitude, higher in threshold, but more persistent. That the higher activation voltage appears on gold despite its films being *thinner*

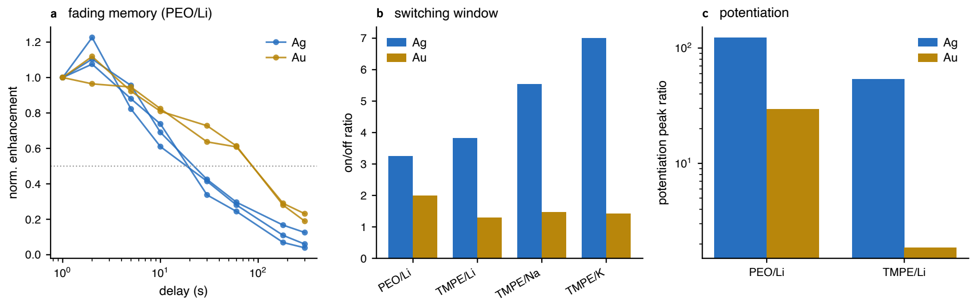

Figure 4.11: Illustrative electrode contrast at the lead composition (0.3/0.09). (a) Fading-memory decays of PEO/LiOTf devices, normalised to the first read: the gold curves (gold) cross half-enhancement later than the silver ones (blue), i.e. gold retains *longer*. (b) The steady-state switching window (on–off ratio) is narrower on gold for every chemistry, and is flat across the TMPE cation series on gold (no Li/Na/K ordering). (c) Pulse-potentiation dynamic range (peak ratio, log scale) is far smaller on gold. The inert electrode thus gives a weaker, higher-threshold, but more retentive response. Small n per cell, no human curve curation on the gold side, and a generation offset between the corpora (see text); analysis in `scripts/ch3_electrode.py`, per-cell values in `handouts/ch3_electrode_by_cell.csv`.

(spun faster) argues against a film-thickness origin and for an interfacial, electrode one [14].

Two of the chemistry conclusions of section 4.5 recur on this second electrode, which is the more useful outcome for a landscape that is otherwise singly sampled. The host ordering persists — PEO retains far longer than TMPE on gold as on silver — and the cation null persists: the gold TMPE/triflate series shows a flat switching window across Li^+, Na^+, and K^+ (on–off $\approx$ 1.3–1.5, no monotone trend), the same absence of ordering the silver retention reported, now seen through a different observable. The gold corpus additionally supplies clean negative controls that the silver set lacked: host-free (semiconductor-plus-salt) and salt-free (semiconductor-plus-host) films show neither a switching window (on–off $\approx$ 1.0) nor potentiation (peak $\approx$ 1), confirming that both the ion-transport polymer and the dissolved salt are required for the memristive response.

These statements are illustrative, and for the familiar reasons. The gold retention rests on two clean devices; the gold curves were not subjected to the per-device visual curation applied to the silver chemistry set (section 4.3); and the corpora differ in fabrication generation (the gold devices are all 2024–2025, the silver chemistry and lead-cell decays 2022–2023), so electrode and generation are partly confounded — the one silver batch fabricated alongside the gold batch, by the same operator, was not measured under the decay protocol. The electrode is therefore reported here as a coherent, mechanistically sensible lever and as cross-electrode corroboration of the chemistry axis, not as a quantitative law; isolating it cleanly would require a single matched-generation batch split between the two electrodes.

4.9 Discussion

The picture that emerges is of a device whose volatile dynamics are controlled by a hierarchy of levers. The strongest and most reliable is the *composition* of the ion-transport phase, which tunes the switching window, the potentiation strength, and the fading-memory time together and in the same direction: more ion-transport polymer yields a weaker, faster-relaxing device. Next come the *host* and *anion* of the electrolyte, which shift the fading-memory time over a wider range still but are documented here only illustratively. The *cation*, which the proof-of-concept chapter's mechanism had suggested would be the controlling variable, is in fact the weakest and least legible lever in this dataset, and its effect is not separable from host, anion, and drive protocol at the available sample sizes. Cutting across all of these is the electrode itself: an inert gold contact, set against the active silver one, weakens the switching and potentiation but lengthens retention (section 4.8), a coarse, mechanistically distinct lever that acts on the electrode interface rather than on the electrolyte chemistry. The hard–soft acid–base framework of **??** is retained here as a qualitative organising idea — it correctly anticipates, for instance, that the most weakly coordinating cation relaxes fastest within one family — but it is explicitly not advanced as a quantitative, transferable law on the strength of these data.

A working mechanistic picture. The composition trends admit a cautious interpretation, offered here as a working hypothesis rather than a demonstrated microscopic mechanism. Increasing the PEO (ion-transport) fraction dilutes and interrupts the *Super Yellow* electronic-percolation pathway, which plausibly accounts for the simultaneous contraction of the switching window and the weaker pulse-driven conductance modulation; the same ion-richer matrix should also let the displaced ionic profile relax more readily, shortening the fading-memory time — so that window, potentiation strength, and retention move together, as observed. The salt fraction, by contrast, plausibly sets the number of mobile, chargeable ionic states available before saturation, which would explain why a higher salt content sustains potentiation to larger pulse counts before turning over. The host and anion effects are most naturally read in terms of cation–ether coordination strength, ion pairing, and segmental mobility [5, 8] — that is, the breadth and depth of the distribution of local activation barriers that underlies the stretched-exponential relaxation [1]. For the ionic-resistance step in this picture there is now direct, independent support: the equilibrium impedance of these films falls along the composition axis and tracks the fading-memory time (section 4.7). Vibrational spectroscopy and X-ray diffraction add further microscopic correlates on the chemistry axis: ATR ties the host's effect to its *local* ether-backbone order (retained in PEO, absent in amorphous TMPE), while XRD shows the composite to be X-ray amorphous in every chemistry — so the order is short-range rather than bulk crystallinity, and the salt is dissolved rather than segregated —

and ATR shows the cation leaving no ion-association fingerprint while UV-Vis shows it leaving the *Super Yellow* optical gap unshifted, consistent with the cation's weak and inconsistent effect on retention and with the chapter's premise that the electrolyte chemistry moves ion transport, not the *Super Yellow* electronic backbone (section 4.5). The remaining links — in particular the anion and barrier-distribution arguments — are not established microscopically here; temperature-dependent measurements and a chemistry-resolved impedance study (section 4.10) would be needed to test them, and the hard–soft acid–base idea is kept strictly qualitative.

The variability that runs through the chapter is a resource, not a defect. The spread of fading-memory times, both across compositions (roughly an order of magnitude) and between nominally identical devices, is precisely the kind of heterogeneity that a physical reservoir requires: a bank of elements with diverse but individually reproducible-enough time constants, whose fading memory of past input is the defining property a reservoir computes upon [15–17]. This is made concrete in fig. 4.12, which places every screened silver device in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane: the PEO fraction sets the broad region a device occupies — low PEO in the long-memory, high-dynamic-range corner; high PEO in the opposite one — so a single composition selects a *paired* operating point, while the substantial scatter within each composition is the genuine device-to-device diversity. **??** takes up this point directly.

4.10 Limitations and Future Work

The principal limitation is one of statistical power off the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices per matched cell, and are therefore illustrative. Four specific consequences should be stated plainly:

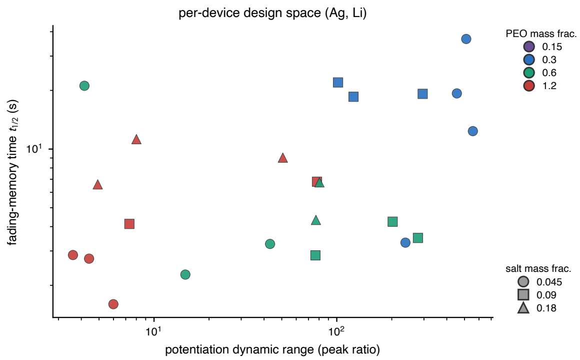

Figure 4.12: Per-device design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by PEO fraction and marked by salt fraction (one point per device, median over its pixels). The PEO fraction selects the broad region a device occupies — low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are not independently tunable but *paired*. The substantial within-composition scatter is the genuine device-to-device heterogeneity that **??** exploits.

1. The cation hypothesis is not tested here at adequate power, and the apparent absence of a $Li^+ > Na^+ > K^+$ ordering should be read as "not demonstrated in this archive" rather than as a positive refutation.

2. The supra-threshold read and the amplitude dependence of the apparent timescale (section 4.6) mean that absolute retention values are protocol-specific.

3. The electrode is now reported as an illustrative axis in its own right (section 4.8) rather than simply held out, but it remains partly confounded with fabrication generation, and device ageing is still an uncontrolled variable that was not modelled.

4. Most consequentially for **??**, pulse integration and forgetting were characterised *separately and at different write amplitudes*: the poten-

tiation curves were acquired at a 3 V write and the fading-memory decays at a 4 V write (table 4.1), at an effectively common inter-pulse interval, and the archive contains no experiment that varies the input timing to probe the interaction of integration and forgetting directly.

Because the drive amplitude itself co-sets the apparent timescale (section 4.6), composing the 3 V potentiation shape with the 4 V decay in **??** assumes the two regimes are compatible — an explicit modelling assumption rather than a measured fact. The clean test is a single matched-amplitude experiment that sweeps both pulse count and inter-pulse interval; it is named as future work.

The corresponding future-work programme is therefore well defined: a dedicated cation series in a single host and anion, at a single fixed protocol and electrode, with a sub-threshold read voltage (as in the proof-of-concept device of **??**) and at least three replicates per cell, would convert the illustrative chemistry landscape into a powered comparison. The impedance corroboration of section 4.7 is likewise extensible: it is reported here only for the replicated PEO/LiOTf composition axis at 0 V DC, whereas the archive also holds bias-resolved spectra (up to 3 V DC) and chemistry-varied devices that were not modelled, so a chemistry- and bias-resolved impedance study would test the host, anion, and barrier-distribution arguments that the present 0 V analysis leaves open. A single matched-generation batch split between silver and gold top electrodes would likewise convert the illustrative electrode contrast of section 4.8 into a clean, generation-controlled comparison. Encapsulation and a systematic ageing study would address the stability dimension.

4.11 Bridge to Temporal Computing

This chapter has shown that the volatile dynamics of *Super Yellow* polymer-electrolyte composites are controllable, and that the control is dominated by composition. Quantitatively, the fading-memory time of the *Super Yellow*/PEO/LiOTf system is tunable over roughly 2 s–20 s by composition alone, with the switching window and potentiation strength varying in concert; chemistry (host, anion) extends this range further, illustratively; and the cation, contrary to the initial hypothesis, is not a robust controlling variable at the available power. A methodological result — that the drive protocol co-sets the apparent timescale — frames how all of this must be measured and compared.

What **??** requires from the present chapter is precisely a set of behavioural ingredients: a composition-indexed family of fading-memory time constants spanning a useful range, a characterisation of how conductance is built up by pulses (including its saturation and turnover), and an account of device-to-device heterogeneity. Those ingredients are now in hand. They also define the shape of the design space the next chapter inherits: the fading-memory time and the input nonlinearity (α) are *coupled* through the PEO fraction (fig. 4.12) — a single composition selects a paired (memory, nonlinearity) operating point rather than two independent settings — while the salt fraction acts as a partially separate lever on the usable dynamic range and turnover. A temporal-computing substrate must therefore be assembled from a *bank* of compositions to span the behaviour it needs, rather than tuned to a single optimum; the heterogeneity catalogued here is what makes that bank possible. The next chapter treats them not as imperfections of a memory but as the raw material of a temporal-computing substrate.

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